Differential-protection probabilistic scoring for methylome-guided ThermoCas9 target-site ranking

Technical memo from the thermocas methylome-analysis framework (audit-trail revision)

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Abstract

Methylation-sensitive Cas9 variants target genomic loci that are hypomethylated in disease cells and methylated in matched normal cells. Selecting such loci from genome-scale array data is a ranking problem whose scoring axis must be cohort-agnostic and must report its own uncertainty. A first-pass probabilistic composite `p_targ × p_prot × p_trust` failed empirically — `p_prot` encoded a static "normal is methylated above 0.5" assumption that is anti-predictive on bulk normal comparators (AUC 0.38). We replace it with a differential factor `p_diff = P(β_normal − β_tumor > δ)` under an independent-normal approximation on per-probe summaries.

This paper is a **scoring-method and benchmarking** contribution, not a target-discovery validation. The headline result is a *rank-lift* finding on a small label set: on the n = 3 validated target probes from Roth et al. (2026) Fig. 5d, scored on Roth's own MCF-7/MCF-10A EPIC v2 cohort (GSE322563), the new composite (V2.5) places all three positives near the top of the ranking (AUC 0.990), compared with raw Δβ-only at 0.974, an uncertainty-aware Δβ_z baseline at 0.940, the deterministic V1 score at 0.821, and the deprecated V2 `tumor_only` mode at 0.928. Δβ-only is itself a strong baseline at this primary endpoint; V2.5′s margin over Δβ-only (+0.016) is small but consistent across the matched cell-line cohorts and label granularities tested (§5.1–§5.2). Because n = 3, AUC is sensitive to the exact negative universe and to the rank of each positive; the manuscript reports per-positive ranks and percentile ranks (§5.1) rather than treating AUC as definitive. A second matched MCF-7/MCF-10A cohort (GSE77348) is the development cohort on which the differential margin δ was tuned and is reported as supporting evidence, not as an independent confirmation. Every benchmark result emits `tie_band_size_at_k` and `precision_at_k_{min,max}` so that low-replicate top-K numbers are reported as adversarial intervals, not point estimates; on n = 2/2 cohorts, V2.5 should

be read as identifying a *top tied candidate class* rather than a ranked top-20. A separate cross-series run at the Sanger GDSC breast panel is documented as an out-of-distribution label-transport boundary case, not a generalization test, because Sanger's MCF-7 is methylated at the sites where Roth's MCF-7 is unmethylated. V1 remains the stable-release default; V2.5 is the recommended probabilistic ranking axis on every non-boundary cohort shape tested (matched cell-line at n = 2/2 and 3/3, and primary tissue at n = 305/50 — §5.1–§5.3), with two caveats: (a) on n = 2/2 cell-line cohorts the visible top-K should be read as a top tied candidate class rather than a ranked shortlist (§6.1); and (b) on primary tissue the shipped defaults σ_floor = 0.05 and δ = 0.2 are not tissue-optimal — §5.3.1 / §5.3.2 show tissue AUC peaks at σ_floor ≈ 0.10 and prefers smaller δ, so tissue-cohort users should run the σ_floor / δ sweep on their cohort before relying on the shipped defaults (regime-specific presets are on the §6.3 follow-up list). All code and benchmark artifacts are public.

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1 · Background

1.1 ThermoCas9 is methylation-sensitive at the PAM

Roth et al. (2026) characterized *Geobacillus thermodenitrificans* T12 Cas9 (ThermoCas9) biochemically, structurally, and in human cell-line editing assays. The central result: cleavage efficiency is governed by the methylation state of the fifth position of the PAM (`5'-NNNN**C**GA-3'` or `5'-NNNN**C**CA-3'`), not by methylation within the protospacer. Methylated 5-methylcytosine (5mC) at the PAM abolishes binding; unmethylated PAMs are cleaved as usual. Cryo-EM structures at 2.2–2.8 Å resolved the protein contacts that discriminate methylated from unmethylated cytosine.

The practical consequence is that ThermoCas9 gives a natural mechanism to selectively edit in cells where a target locus is hypomethylated while sparing cells where the same locus is methylated. Roth demonstrated this in MCF-7 vs. MCF-10A at three breast-cancer loci: *EGFLAM* (control locus included in the validated target set), *ESR1*, and *GATA3*. The underlying methylation of the PAM cytosine at those loci was profiled on the Illumina Infinium MethylationEPIC v2 array and reported in Supplementary Figure 5d of the main paper alongside the editing results.

1.2 Target-site selection is a ranking problem

Given a reference genome, the ThermoCas9-compatible PAM sites on any chromosome number in the millions. To pick candidates for wet-lab follow-up, one wants a ranked list where the top scorers are *biologically coherent* — meaning the candidate's PAM cytosine is unmethylated in the disease cohort and methylated in the matched normal — *and* where the confidence of the underlying methylation measurement is part of the score. This is a classical ranking-with-covariates problem, and the natural scoring target is some decomposition of a probability of "therapeutic selectivity."

This paper describes the specific decomposition we ended up using after two earlier attempts failed on empirical cohorts.

1.3 Notation

For one candidate site on one cohort we carry six per-probe β summary statistics from the cohort's methylation array (`β_tumor_mean`, `β_tumor_q25`, `β_tumor_q75`, `β_normal_mean`, `β_normal_q25`, `β_normal_q75`), per-side sample counts (`n_tumor`, `n_normal`), and an `EvidenceClass` capturing the distance between the candidate's PAM cytosine and the nearest assayed CpG probe (`EXACT`/`PROXIMAL_CLOSE`/`PROXIMAL`/`REGIONAL`/`UNOBSERVED`) — nine inputs in total per candidate × cohort pair.

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2 · The V2 misspecification

2.1 First-pass composite

The first-pass composite followed a standard decomposition:

`p_therapeutic_selectivity = p_targ × p_prot × p_trust`

- `p_targ = P(β_tumor < 0.30)` — probability that the tumor's PAM cytosine is unmethylated enough for ThermoCas9 to cleave. Estimated via a method-of-moments Beta fit to `(mean, IQR/1.349)` with the regularized incomplete beta computed in pure stdlib via Lentz's continued fraction; falls back to a piecewise-linear CDF when the Beta moments are ill-defined.
- `p_prot = P(β_normal > 0.50)` — probability that the normal's PAM cytosine is methylated enough to protect the normal cell.
- `p_trust` — confidence factor that scales with `EvidenceClass` (0.95 at EXACT, 0.15 at REGIONAL, 0.0 at UNOBSERVED) and saturates linearly in `min(n_tumor, n_normal)` up to `ramp_n = 30`.

2.2 Empirical failure of `p_prot`

We ran a factor ablation on a structural surrogate for the Roth MCF-7 vs. MCF-10A comparison: GEO series GSE77348, an independent 2016 breast cell-line methylation series on HM450 (not the Roth samples). With 1,687 gene-region-filtered "positives" at the Roth genes, the measured AUCs were:

axis	AUC (loose)	AUC (tight)
`v2_tumor_only` (p_targ × p_trust)	0.733	0.770
`p_targ` alone	0.683	0.717
V1 `final_score`	0.657	0.628
`naive_selectivity` (β_n − β_t)	0.608	0.571
`p_targ × p_prot`	0.503	0.488
`p_prot` alone	**0.384**	**0.343**

`p_prot` is anti-predictive. Multiplying it against `p_targ` destroys the signal. The cause is the semantics: `P(β_normal > 0.5)` assumes that the normal comparator *methylates the target*. That assumption holds for Roth's specific MCF-10A cell line at the three *ESR1* / *GATA3* / *EGFLAM* probes they chose to validate (genes that are silenced in non-tumorigenic mammary epithelium) but does not generalize to bulk adjacent-normal cohorts or even to MCF-10A at gene-body probes.

2.3 V2.4 — keep the composite, drop `p_prot` by default

The intermediate fix kept the composite but allowed users to opt in or out of the anti-predictive factor. Default changed to `tumor_only` = `p_targ × p_trust`. This improved global AUC but the top-100 collapsed onto "always-unmethylated" loci (candidates where both tumor *and* normal are low-methylated) — good AUC, unusable for target discovery. The deeper fix was needed.

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3 · V2.5 · differential-protection probabilistic mode

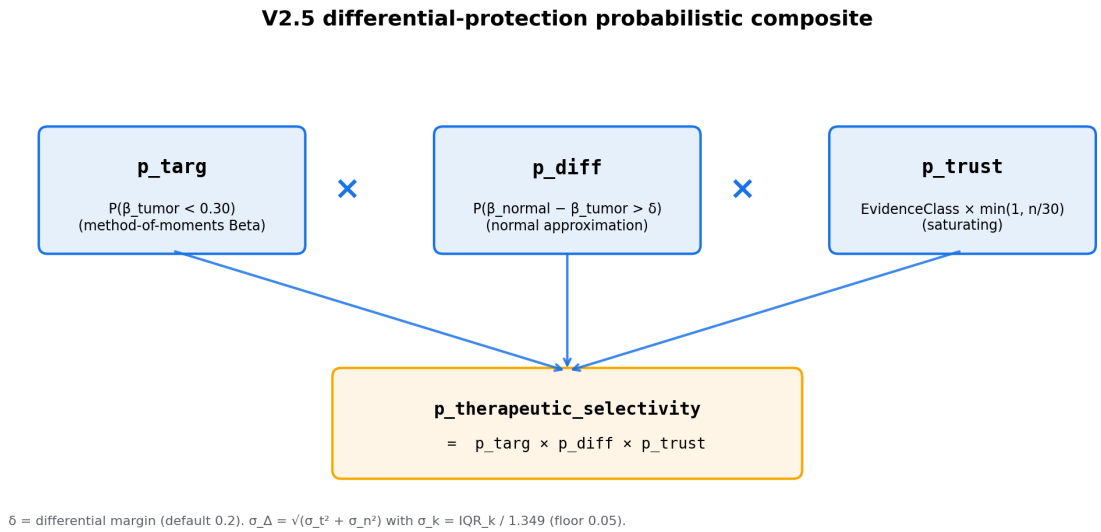

Figure 1. The V2.5 composite. Three factors multiply: `p_targ` (tumor unmethylated at the PAM cytosine), `p_diff` (differential methylation gap exceeds δ under a normal approximation on per-probe β summaries), and `p_trust` (evidence-class confidence, saturating in min sample count). The result is `p_therapeutic_selectivity`, the stored scalar that downstream ranking consumes. The deprecated V2 `p_prot = P(β_normal > 0.5)` is replaced by `p_diff` to remove its static-threshold assumption about the normal side.

3.1 Formulation

Replace the threshold-based `p_prot` with a differential factor:

```
p_diff(δ) = P(β_normal − β_tumor > δ)
```

estimated under an independent-normal approximation:

```
σ_k   ≈ IQR_k / 1.349                (k ∈ {tumor, normal}; floor at 0.05)
σ_Δ²  = max(σ_t, floor)² + max(σ_n, floor)²
z     = (δ − (μ_n − μ_t)) / σ_Δ
p_diff = 1 − Φ(z) = 0.5 (1 − erf(z / √2))
```

with Φ the standard normal CDF. The composite becomes:

```
p_therapeutic_selectivity = p_targ × p_diff × p_trust
```

This makes no absolute-threshold claim about the normal side. It asks only what fraction of normal-vs-tumor β pairs sampled from the per-side empirical distributions would exceed δ, under independent normal approximations to those per-side distributions. (This is a *population-overlap* statistic, not a confidence interval on a cohort-level mean contrast — see §3.3.)

3.2 Choice of δ

An offline δ sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 surrogate selected δ = 0.2 by joint optimization of AUC and `P@100` against the *pre-repair* (gene-universe) positives that were in use at the time of selection. Under that pre-repair tuning objective, larger δ values monotonically reduced both metrics — the factor was too strict to admit candidates with moderate but real differentials.

The post-repair sensitivity story is more nuanced and is reported in §5.3.2: against the Roth-validated label set, δ = 0.3 (and on some cohort × tier rows, δ = 0.4) marginally outperforms δ = 0.2 on the matched cell-line cohorts; tissue (GSE69914) prefers smaller δ. We do not retune δ post hoc against the post-repair labels, because doing so would invalidate the dev-cohort separation discipline of §4.0. δ = 0.2 is therefore retained as the shipped default; readers wanting the full sensitivity surface should consult §5.3.2.

The default is exposed as `differential_delta` in the cohort configuration. The δ used is recorded on every emitted `ProbabilisticScore` record so the margin in effect is recoverable from a single output row.

3.3 What `p_diff` is a probability over

`p_diff(δ) = P(β_normal − β_tumor > δ)` is a probability under the *biological β-heterogeneity* implied by per-cohort, per-probe quartiles — not a posterior probability of editing efficacy and not a confidence interval on a cohort-level mean contrast. The distribution being integrated is the one that would generate single-sample β values from a tumor-side population with quartiles (`β_tumor_q25`, `β_tumor_q75`) and a normal-side population with quartiles (`β_normal_q25`, `β_normal_q75`), under independent normal approximations on each side. Two consequences follow:

1. **The variance does not shrink with sample size.** Because the per-side σ is the IQR-derived population dispersion, not the standard error of a mean, `p_diff` does not become more confident as `n_tumor`/`n_normal` grow. The growing-`n` story for the composite is carried by `p_trust` (saturating at `n ≳ 30`), not by `p_diff` tightening.
2. **`p_diff` is therefore a population-overlap statistic, not a significance test.** It answers "what fraction of tumor-vs-normal β-pairs sampled from the cohort would exceed δ?" An alternative formulation that scales σ with `n` (e.g. SE on the mean difference, or a hierarchical model on per-side variance) would also be defensible and would give a different — and at low `n`, a much wider — distribution. We use the population-overlap form because (a) the comparison the user cares about for site selection is biological — "is this site differentially methylated *between cells*?" — not "does this cohort have enough power to call this site significant?"; and (b) the small-`n` power question is already absorbed by `p_trust`. Re-deriving the composite under an SE-on-mean variant is a documented future axis (§6.3).

This is a deliberate modeling choice and is the single most common point of confusion when reading `p_diff` cold.

3.4 Why a normal approximation instead of a difference-of-Betas

A proper Bayesian treatment would model β-values as Beta distributions and compute the exact difference distribution numerically. For the IQR ranges seen on HM450 / EPIC v2 arrays (~0.05–0.20), the normal approximation is close to the correct answer and has the advantage of being a two-line stdlib closed form. A σ floor of 0.05 prevents σ_Δ from collapsing to zero at boundary β-values (very common for CpG islands methylated at 0 or 1). These tradeoffs are documented in the code; a downstream user who wants the exact difference-of-Betas path can swap it in.

The normal approximation is operationally adequate for the *ranking* use case but is not a calibrated probability of editing success. Two situations where the approximation is poor on its own terms but does not corrupt the ranking:

- **Boundary-β pile-ups.** β-distributions at CpG islands often pile up near 0 or 1; a single-mode normal centered at the per-side mean overstates symmetry and understates skew. The σ floor blunts the worst case (σ_Δ ⟶ 0) but does not restore the correct shape.
- **Bimodality across samples.** When tumor or normal samples carry two β sub-populations (e.g. methylation-class heterogeneity in primary tissue), the IQR-derived σ averages the two modes; the per-side

normal then sits between the modes rather than on either. `p_diff` in this regime under-states the true population overlap.

Neither failure mode is silent: the §5.5 annotation pipeline flags candidates whose per-side β quartiles indicate strong bimodality. A future axis based on the exact difference-of-Betas or a two-component mixture would address both.

3.5 The low-n tied-band prediction

With the new factor isolated, we can predict before running any new cohort. `p_trust` is `EvidenceClass.base × min(1, min(n_t, n_n) / 30)`: piecewise-linear in min sample count up to `ramp_n = 30`, then a constant ceiling per evidence class. It is **never** continuous-valued in the real-line sense — at n ≥ 30 it saturates to a discrete-by- EvidenceClass value (0.95 at EXACT, 0.75 at PROXIMAL_CLOSE, etc.). What changes between the low-n and high-n regimes is which factor *dominates* the composite ordering:

- **Low n (n ≪ 30).** `p_trust` scales every same-evidence-class candidate by the same uniform fraction `(n / 30)`. Per-side IQRs are wide (or zero, hitting the `σ_floor`), pushing both `p_targ` and `p_diff` toward saturation (≈1) for any candidate with clean tumor hypomethylation and a large normal/tumor gap. The composite is effectively `1 × 1 × p_trust` for that whole subset, producing a large tied band at the top of the score distribution.
- **High n (n ≥ 30).** `p_trust` is constant per EvidenceClass. Per-side IQRs are realistic (HM450 ranges of 0.05–0.20). The composite ordering is now driven by the continuous variation in `p_targ × p_diff`, with `p_trust` only multiplying by a per- evidence-class constant. The previously-tied subset spreads out by its underlying `p_targ × p_diff` values.

The falsifiable prediction: the K = 100 tied band on V2.5 should be large at n=2/3 per side and shrink to near-1 at n ≳ 30 per side. §5.1 and §5.3 test this directly across cohorts ranging from n=2/2 to n=305/50.

The K = 100 large-band regime is what motivates the "top tied candidate class" framing in §6.1: on n=2/2 cohorts the visible top-20 is a 20-record window inside a 190- to 421-record tied band, not a stable rank ordering. Per-positive ranks (§5.1) and AUC (§5.1, §5.3) are the stable claims at low `n`.

Quartile convention on n=2 records. Per-side β quartiles are computed with Python's `statistics.quantiles(..., method="inclusive")` (R-7 quantile definition) and hard-clamped to the observed sample range at every probe, so q25 / q75 / mean are always within [min(β_samples), max(β_samples)]. On n = 2 samples per side this is deterministic (IQR = 0.5 × |β_max − β_min|) but weak as a population-dispersion estimate: the "IQR" it produces is a function of two observations, not a quantile of an underlying distribution. Because the `σ_floor = 0.05` kicks in whenever IQR/1.349 < 0.05 (i.e. whenever the two observations differ by less than ~0.135), σ_floor dominates σ_Δ on most n = 2 records.

Quantified σ_floor binding rate per cohort (reproducible via `uv run scripts/sigma_floor_binding_rate.py`; denominator is records with both β means present):

cohort	σ_floor binds on either side	σ_floor binds on both sides
GSE322563 HM450 (n = 2/2)	**100.0%**	**99.9%**
GSE322563 native (n = 2/2)	**100.0%**	**99.9%**
GSE77348 (n = 3/3)	**100.0%**	**99.5%**
GSE69914 (n = 305/50 tissue)	65.1%	43.2%

On matched cell-line cohorts the σ_floor is the binding constraint on essentially every observed record (both sides). On tissue the true IQRs are wide enough that the σ_floor binds on roughly two thirds of records either-side, and on both sides for only ~43%. This is the mechanistic origin of both the low-`n` tied band (§5.3.1) and the §5.3.2 finding that tissue AUC prefers a larger σ_floor: on cell-line cohorts, σ_floor is determining σ_Δ almost everywhere, so changing its value re-shapes the whole composite.

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4 · Benchmark methodology

4.0 Analysis plan: primary endpoint, sensitivity, boundary

To guard against post-hoc optimization, the analysis design separates tuned components from evaluation. The hyperparameter `differential_delta` was tuned on a specific cohort, and that cohort is therefore *not* part of the protected primary endpoint. The plan below is not a formal pre-registered protocol with a third-party timestamp; it is an analysis plan whose ordering is independently verifiable from the git history (commit hashes given inline).

- **Tuned, fixed before cross-cohort evaluation.** `differential_delta = 0.2` was selected by an offline sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 MCF-7/MCF-10A surrogate against the *pre-repair* (gene-universe) positives. The value was fixed in code at commit `aece3de` ("V2.5: experimental differential-protection probabilistic mode (opt-in)", 2026-04-20), *before* the GSE322563 (Roth) cohort was ingested or benchmarked. GSE77348 is the **development cohort** for δ; results on GSE77348 are tuned-on supporting evidence, not independent confirmation. The label-repair exercise that produced the Roth-validated positives is data preparation, not scoring-axis tuning, and it post-dates the δ choice.
- **Primary endpoint.** AUC at the `positives_roth_validated.txt` label set (n = 3 Roth Fig. 5d target probes, hg38 ⟶ hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples). This is the single independent paper-comparable test. δ was not tuned against this cohort; the label set was not derived from it. Because n = 3, AUC is a *rank-lift* statistic on the three known positives — read alongside the per-positive ranks and percentile ranks reported in §5.1, not as a generalized discovery claim.
- **Supporting evidence** (§5.1, §5.2). AUC on GSE77348 (the development cohort for δ, MCF-7/MCF-10A on HM450). Same matched cell-line biology as GSE322563 but profiled by a different lab on a different platform, so cross-platform agreement on the V2.5 ordering is informative — but it is *not* an independent cohort under the analysis-plan discipline because δ was selected on it.
- **Sensitivity analyses** (§5.2). The `narrow` (±50 bp) and `wide` (±500 bp) label sets are alternative label granularities; the cross-granularity AUC ordering stability is reported without retuning δ. `P@K` intervals on tied bands are reported but are expected to be uninformative on low-`n` cohorts (§3.5).
- **Tissue-cohort behavior** (§5.3). GSE69914 (n = 305 / 50 primary breast tissue) is *not* a paper-comparable test of Roth biology — it probes whether the V2.5 architecture's low-`n` tied-band behavior dissolves at high `n`, as predicted in §3.5.
- **Out-of-distribution boundary case** (§5.4). GSE68379 (Sanger GDSC breast panel × GSE69914 healthy normal) is *not* a generalization cohort; it is a label-transportability test. It is reported separately and is not pooled into any primary-endpoint average.

4.1 Cohorts

Four cohorts, all public HM450 or EPIC v2 methylation arrays:

cohort	regime	n tumor / normal	platform	role
GSE322563	Roth actual samples	2 / 2	EPIC v2	paper-comparable biology
GSE77348	MCF-7/ MCF-10A surrogate (DNMT3B paper)	3 / 3	HM450	δ development cohort (tuned-on supporting evidence)
GSE69914	primary breast tissue	305 / 50	HM450	high-n tissue validation
GSE68379	Sanger GDSC cell-line panel × GSE69914 healthy normal	52 / 50	HM450	orthogonal (see §5.4)

Ingest details:

- **GSE322563.** Supplementary β matrix `GSE322563_beta_matrix_EPIC_v2.txt.gz`. EPIC v2 probe IDs carry `_BC##` / `_TC##` / `_TO##` / `_BO##` beadchip suffixes; we strip to the canonical `cg*` identifier and intersect with HM450 — 80.7% retention. Roth-positive gene retention: ESR1 96%, EGFLAM 91%, VEGFA 90%, GATA3 83%. The main paper's data-availability statement prints `GSE32256`, which resolves to a *Paramecium* transcriptome series; the supplement's reporting summary correctly prints `GSE322563`. We verified the liftover by cross-checking our per-probe β values at the three Roth-Fig.-5d target positions against the paper's published β — every value agrees to two decimal places (EGFLAM 0.01/0.49; ESR1 0.07/0.94; GATA3 0.02/0.31).
- **GSE77348.** Per-sample β values computed from Unmethylated / Methylated intensities (β = M / (M + U + 100)); restricted to untreated replicates.
- **GSE69914.** Full SeriesMatrix β table; filtered to `status == 2` (sporadic breast cancer, non-BRCA1) for tumor and `status == 0` (healthy donor, non-BRCA1) for normal. Tumor-adjacent normals and BRCA1 carrier arms excluded.
- **GSE68379.** Sanger processed matrix has integer row indices; we rebuild cg probe IDs via the Illumina GPL13534 manifest (485,577 data rows; row N → IlmnID N); filtered to 52 primary-site-breast cell lines. No in-study normal — paired cross-series with GSE69914 healthy normals.

All cohort builder scripts are in `scripts/build_*.py`. Every run produces `tumor_summary.tsv` + `normal_summary.tsv` + `PROVENANCE.md` in the shared `LocalSummaryBackend` format; the scorer is platform-agnostic thereafter.

4.2 Positives (label repair)

Our first benchmark used "gene-universe" positives: HM450 probes in the neighborhood of Roth's validated gene symbols (~970 tight, 1,687 loose, on chr5/6/10). A diagnostic on GSE322563 revealed this label set is noisy: only 23% of Roth-gene probes have β_n − β_t > 0.2 on the actual Roth samples. Most are gene-universe members, not validated target loci.

We rebuilt the positives from Roth Supplementary Fig. 5d which names three exact MCF-7 / MCF-10A validated target coordinates in hg38. These were lifted to hg19 via Ensembl REST `/map/human/GRCh38/.../GRCh37`:

target	gene	hg38	hg19 (lifted)	β shift (hg19 vs hg38)
EGFLAM T11	EGFLAM	chr5:38,258,842	chr5:38,258,944	+102 bp
ESR1 T17	ESR1	chr6:151,690,043	chr6:152,011,178	+321,135 bp
GATA3 T18	GATA3	chr10:8,045,425	chr10:8,087,388	+41,963 bp

Three positives files result, at increasing granularity:

- `positives_roth_validated.txt` (n = 3): closest NNNNCGA candidate per Roth target.
- `positives_roth_narrow.txt` (n = 28): NNNNCGA candidates within ±50 bp of each Roth target.
- `positives_roth_wide.txt` (n = 142): NNNNCGA candidates within ±500 bp of each Roth target.

4.3 Score axes

Four axes are benchmarked on every cohort:

- **Δβ-only** (`naive_selectivity`) — `β_normal_mean − β_tumor_mean`. The literature-naive baseline: rank by the raw methylation gap with no uncertainty propagation or evidence weighting. Reported up front so the value of the additional factors is auditable against the simplest possible scoring axis.
- **V1** — `final_score = sequence × selectivity × confidence − heterogeneity_penalty − low_coverage_penalty`. Deterministic, continuous-valued, tie band always 1.
- **V2 `tumor_only`** — `p_targ × p_trust`. Retained for audit; default in v0.4.0.
- **V2.5 `tumor_plus_differential_protection`** — `p_targ × p_diff × p_trust` with $\delta = 0.2$.

4.4 Platform harmonization and catalog scope

Two facts constrain interpretation of the reported numbers and belong in the methodology, not as a footnote:

- **Platform.** GSE322563 was profiled on Illumina EPIC v2 (GPL33022, ~937K probes); GSE77348, GSE69914, and GSE68379 are on HM450 (GPL13534, ~485K probes). The primary GSE322563 results in §5.1 use an HM450-intersect harmonization path (EPIC v2 probe IDs stripped of beadchip-design suffixes — `_BC##`, `_TC##`, `_TO##`, `_BO##` — then intersected with the HM450 universe; 80.7% retention overall, 83–96% at the Roth-validated genes). To quantify the cost of that shortcut, we also built a **native EPIC v2 ingest** (script `build_epic_v2_probes.py` lifts GPL33022 hg38 ⟶ hg19 via pyliftover; 147,928 chr5/6/10 probes, 99.95% liftover success; cohort builder `build_gse322563_native_epic_v2_cohort.py` keeps native probe IDs and produces a 5.22M-candidate catalog vs the HM450 catalog's 2.98M). Side-by-side at the validated label set on GSE322563: V2.5 differential AUC 0.990 (HM450-intersect) ⟶ 0.986 (native EPIC v2), a 0.003 difference well within the cohort's tied-band noise floor. V1 gains the most under the native ingest (+0.05 to +0.11 AUC across label granularities) because its continuous score ranks the validated targets higher relative to easy-negative candidates added by the EPIC v2 probe density. The only axis ordering that holds on **every** matched-cell-line tier × path row (9/9) is V2.5 > V1 and V2.5 > Δβ-only; the relative order of Δβ-only, V1, and V2 `tumor_only` below V2.5 reshuffles across rows (e.g. `tumor_only` sits above Δβ-only on both GSE322563 wide tiers; on GSE77348 `tumor_only` is the lowest at every tier). Sensitivity table in §5.2; benchmark JSONLs at `examples/gse322563_native_roth_labels/`.
- **Catalog scope.** The candidate catalog is chr5 / chr6 / chr10 only, filtered to candidates within ±500 bp of any assayed HM450 probe (~2.98M NNNNCGA / NNNNCCA candidates). The three chromosomes were

chosen because they carry the Roth Fig. 5d validated targets (EGFLAM on chr5, ESR1 on chr6, GATA3 on chr10) and because they provide a realistic but not whole-genome test bed. All AUC / `P@K` / tie_band numbers in §5 are on this scope. Extending to whole-genome hg19 is an infrastructure change, not a scientific one.

4.5 Tie-band handling

Earlier iterations of the benchmark sorted candidates by score alone, with input-order as the implicit tie-break. On cohorts with low n, the score distribution's top can sit inside a tied band of 190–11,000 records, making the reported `P@K` a function of serialization order rather than of the scorer.

The shipped benchmark contract:

1. Sort primary by score descending, secondary by `candidate_id` ascending (`evaluate_ranking`'s explicit `tie_break_policy = "candidate_id_asc"`).
2. Report `tie_band_size_at_k`: number of records tied at the K-th position's score. When > 1, top-K membership is partially determined by the secondary key.
3. Report `precision_at_k_min` and `precision_at_k_max` (plus recall counterparts) — adversarial tie-break bounds over the tied band. The worst-case pushes every tied-band positive out of top-K; the best-case pulls them in. When `tie_band_size_at_k == 1` both bounds collapse to the observed `precision_at_k`.

These three fields are on every emitted `BenchmarkResult` JSONL row.

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5 · Results

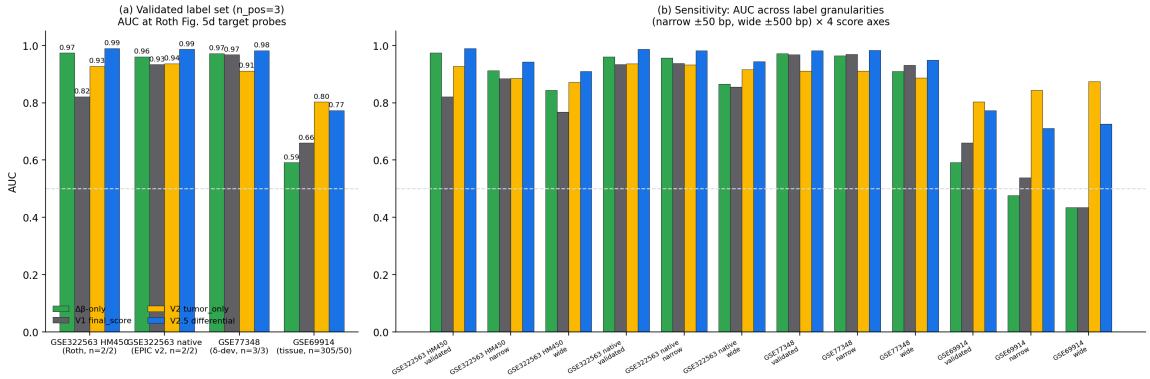

Figure 2. Cross-cohort AUC. **(a)** Validated-label AUC on each cohort. GSE322563 is the independent primary endpoint; GSE77348 is the development cohort on which δ was tuned (§4.0) and is shown as supporting evidence; GSE69914 is the tissue-behavior check for the §3.5 tied-band prediction. V2.5 is the highest-AUC axis on both matched cell-line cohorts; on tissue `tumor_only` takes a small AUC lead but at a tied band of 6,540 (§5.3 + Fig. 3). V1 falls below random on the tissue wide-label set in (b). **(b)** Sensitivity over label granularity (validated ⟶ narrow ±50 bp ⟶ wide ±500 bp). **V2.5 remains the highest-AUC axis at every (matched cell-line cohort × granularity) cell**; the ordering among the lower axes (V1, `tumor_only`, Δβ-only) reshuffles between tiers — e.g. `tumor_only` overtakes Δβ-only at GSE322563 wide, and V1 overtakes Δβ-only at GSE77348 narrow / wide. The only stable ordering claim is V2.5 on top. Dashed line at AUC = 0.5 (random). The out-of- distribution GSE68379 cohort is not included here — it is plotted separately (§5.4).

5.1 Primary endpoint — matched cell-line AUC at validated Roth probes

Primary endpoint (independent): AUC at `positives_roth_validated.txt` (n = 3 Roth Fig. 5d target probes, hg38 ⟶ hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples, EPIC v2). δ was not tuned on this cohort; the label set was not derived from it.

Supporting evidence (development cohort for δ): AUC on **GSE77348** (MCF-7 / MCF-10A on HM450, the DNMT3B-paper surrogate). Same matched cell-line biology, different platform and laboratory. Reported here because same-axis ordering on a second lab's MCF-7/MCF-10A is informative, but with the caveat that δ was selected on this cohort (§4.0) so its AUC is tuned-on, not independent.

axis	**GSE322563** (primary, n_pos=3, n_neg=2,979,994)	GSE77348 (δ-tuned, n_pos=3, n_neg=2,979,994)
Δβ-only (`β_n − β_t`)	0.974	0.972
Δβ_z (uncertainty-aware Δβ)	0.940	0.956
V1 `final_score`	0.821	0.968
V2 `tumor_only`	0.928	0.912
V2.5 differential	**0.990**	**0.982**

Reading the axis grid. Δβ-only (raw mean gap) is already a strong baseline — it lifts the validated positives to AUC 0.974 on the primary cohort. The uncertainty-aware Δβ_z baseline (`(β_n − β_t) / sqrt(σ_t² + σ_n²)`, using the *same* IQR-derived σ as V2.5's `p_diff` denominator; reproducible via `uv run scripts/delta_beta_z_baseline.py`) lifts to AUC 0.940 — slightly *below* raw Δβ-only because the σ-based denominator penalizes positives with wide IQRs. V2.5 adds **+0.016** over Δβ-only and **+0.050** over Δβ_z on the primary endpoint; on GSE77348 the corresponding margins are +0.010 and +0.026. These are small but consistent across the matched cell-line cohorts and label granularities tested (§5.2). The `p_targ` (tumor-side unmethylation) and `p_trust` (evidence / sample saturation) factors of V2.5 are therefore pulling real weight on top of the per-side dispersion signal — V2.5 is not just a re-skinned Δβ_z — but the honest framing is that the shipped baselines (especially raw Δβ-only) already do most of the rank-lift work on matched cell-line cohorts; V2.5's additional contribution is the tie-band-honest top-K reporting (§4.5) and the probability-scale composition with future factors, not the AUC point alone.

On the independent primary endpoint (GSE322563 validated), V2.5 is the highest-AUC axis — +0.17 over V1 and +0.06 over the deprecated V2 `tumor_only` composite. On the development cohort (GSE77348 validated), V2.5 is similarly the highest-AUC axis (+0.01 over V1, +0.07 over `tumor_only`); this is consistent with the primary- endpoint result but is not an independent replication, since the same cohort selected δ = 0.2 pre-repair-labels (§4.0).

5.1.1 Per-positive ranks of the n = 3 validated targets

AUC summarizes the rank lift across the three validated positives, but with `n_pos = 3` it is highly sensitive to the rank of each one. The table below gives the per-positive rank, percentile rank, score, and per-side β values on each cohort × axis. (Full grid including GSE69914, all four axes including Δβ-only, is at `examples/validated_positive_ranks.md`; reproduce with `uv run scripts/validated_positive_ranks.py`.)

GSE322563 HM450 (primary, N=2,979,997 candidates):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5 rank (%ile)	Δβ-only rank (%ile)
ESR1	0.07	0.94	0.87	**844** (99.97%)	**2,575** (99.91%)	4,742 (99.84%)
EGFLAM	0.02	0.49	0.48	839,106 (71.84%)	64,433 (97.84%)	67,620 (97.73%)
GATA3	0.03	0.31	0.28	755,885 (74.64%)	24,083 (99.19%)	159,932 (94.63%)

GSE77348 (development cohort for δ, N=2,979,997):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5 rank (%ile)	Δβ-only rank (%ile)
ESR1	0.03	0.87	0.84	**1,618** (99.95%)	**3,965** (99.87%)	8,733 (99.71%)
EGFLAM	0.005	0.34	0.33	257,041 (91.37%)	124,925 (95.81%)	178,663 (94.01%)
GATA3	0.015	0.59	0.58	31,061 (98.96%)	**28,325** (99.05%)	61,254 (97.94%)

Reading the table. *ESR1* drives the AUC story: it is in the top ~0.05% of the score distribution under V2.5 on both matched cell-line cohorts, with V1 also placing it well. *EGFLAM* and *GATA3* sit in the top 1–3% under V2.5 — meaningful rank lift over V1, but **outside the top 100 on every axis tested**. None of the three validated positives appears in any cohort's top-20 (§5.5). This is the operational meaning of "rank lift on n=3": V2.5 reliably places known positives in the top percentile of millions of candidates, but on n=2/2 cohorts the top tied band at K=100 is large enough that the validated positives sit *behind* a 190- to 421-record block of equally-scored candidates.

The ESR1-dominance also implies AUC is fragile: if ESR1 alone moved out of the top 10,000 on GSE322563, the AUC would drop from 0.990 to roughly the average of the other two positives' percentiles ≈ 0.985 (still high), but the AUC *gap* between V2.5 and Δβ-only would also compress.

5.1.2 Descriptive AUC uncertainty

At `n_pos = 3`, exact / permutation / bootstrap AUC intervals can be defined but are coarse and dominated by the rank of each positive — they are not very informative or stable as inferential statistics. We therefore report two descriptive distributions instead (reproducible via `uv run scripts/auc_uncertainty.py`):

- **Permutation null** — 10,000 draws of three random candidates uniformly without replacement from the full candidate set; AUC computed under benchmark mid-rank tie handling. The empirical *p*-value is `Pr(AUC_random_triple ≥ AUC_observed)`.
- **Negative bootstrap** — 1,000 resamples of the negative set with replacement (the three positive scores held fixed). The bootstrap spread is small by construction here because we hold the positives' ranks fixed; it is reported for completeness.

cohort	axis	observed AUC	random-triple null 2.5–97.5%	one-sided p (Pr ≥ obs)
GSE322563 HM450	V1 final_score	0.821	[0.166, 0.815]	0.021
GSE322563 HM450	Δβ-only	0.974	[0.179, 0.821]	2×10^{-4}
GSE322563 HM450	**V2.5 diff**	**0.990**	[0.338, 0.798]	**$\leq 1 \times 10^{-4}$**
GSE322563 native v2	V1 final_score	0.933	[0.206, 0.819]	2×10^{-3}
GSE322563 native v2	Δβ-only	0.961	[0.173, 0.819]	6×10^{-4}
GSE322563 native v2	**V2.5 diff**	**0.986**	[0.313, 0.811]	**$\leq 1 \times 10^{-4}$**
GSE77348	V1 final_score	0.968	[0.261, 0.821]	3×10^{-4}
GSE77348	Δβ-only	0.972	[0.175, 0.821]	3×10^{-4}
GSE77348	**V2.5 diff**	**0.982**	[0.176, 0.821]	**$\leq 1 \times 10^{-4}$**

What this table does and does not say:

- **It shows that none of the four axes reach their observed AUC by chance.** The one-sided p is ≤ 0.021 on every cohort × axis cell. The rank-lift signal at `n_pos = 3` is real for every axis tested, not just for V2.5.
- **V2.5′s p is at the resolution floor on all three cohorts.** V2.5 is $p \leq 1 \times 10^{-4}$ on every cohort (the resolution floor of 10,000 draws). On GSE322563 HM450 this is at least two orders of magnitude stricter than V1 (`p = 0.021`); on native EPIC v2 and GSE77348 the gap is smaller (V1 `p = 0.002` and 3×10^{-4} respectively), so the matched cell-line result is cleaner on the HM450 path than on the other two. A finer-grained null (e.g. 10^6 draws) would be needed to distinguish V2.5 from the other strong axes here; with a 10^4-draw null several axes sit at the resolution floor.
- **This is an against-random check, not a superiority test between axes.** The null asks "does this axis separate these three positives from random triples?" Direct pairwise comparison between V2.5 and Δβ-only on the same positives is reported separately in §5.1.3 and is correctly framed there as descriptive, not inferential (sign-flip `p_one_sided = 0.125` at `n_pos = 3`).

We do **not** report the negative-bootstrap spread in the body — because positive scores are held fixed and the negative pool is 3×10^6 records, the spread collapses to ±0.001 across every cell (see `examples/auc_uncertainty.md`). At this scale the negative pool is essentially a population, not a sample, so the bootstrap spread is mechanical rather than inferentially meaningful.

5.1.3 Paired per-positive rank lift: V2.5 vs Δβ-only

The §5.1.2 permutation null tests each axis separately against uniformly-random 3-record positive triples. That answers whether each axis separates *these three* positives from the rest of the candidate set, but does not directly compare V2.5 against Δβ-only *on the same positives and same negative universe*. The direct paired comparison (reproducible via `uv run scripts/paired_rank_lift_v25_vs_dbeta.py`):

cohort	*ESR1* Δβ-only rank / V2.5 rank	*EGFLAM* Δβ / V2.5	*GATA3* Δβ / V2.5	positives where V2.5 > Δβ-only
GSE322563 HM450	4,742 / 2,575	67,620 / 64,433	159,932 / 24,083	**3 / 3**
GSE322563 native v2	9,085 / 5,746	185,641 / 158,502	422,569 / 50,096	**3 / 3**
GSE77348	8,733 / 3,965	178,663 / 124,925	61,254 / 28,325	**3 / 3**

On every matched cell-line cohort, V2.5 ranks **all three** validated positives higher than Δβ-only does. The aggregate rank lift (sum over the three positives of `r_Δβ − r_V2.5`) is +141,203 (HM450), +402,951 (native EPIC v2), and +91,435 (GSE77348) — V2.5 moves each positive multiple thousand positions up the ranking on average.

Where V2.5′s paired advantage concentrates. The per-positive table makes a secondary observation visible: *GATA3* — the validated positive with the smallest raw β gap (Δβ ≈ 0.28 on GSE322563 HM450, vs ≈ 0.48 for EGFLAM and ≈ 0.87 for ESR1; §5.1.1) — is where V2.5′s advantage over Δβ-only is largest on every cohort (+135,849 positions on HM450, +372,473 on native, +32,929 on GSE77348; cf. +2k–+27k for the other two positives). This is consistent with the intended role of `p_targ × p_trust`: they help most where the naive gap signal is weakest, not where it is already large. One positive is not a strong basis for a general claim, but the direction is the one the composite was designed to produce.

Statistical caveat. A paired sign-flip permutation null ($2^3 = 8$ equiprobable sign patterns, each flipping independently which axis is "assigned" to which per-positive rank) floors the one-sided *p*-value at **0.125** when all three signs agree — which they do on every cohort here. With `n_pos = 3` this is the best resolution the paired test can achieve. Read it as a *descriptive* 3-for-3 agreement with the direction of the V2.5 AUC improvement, not as an inferential significance result.

5.2 Sensitivity analyses: label granularity and P@K intervals

Label granularity (AUC). Stability of the primary-endpoint ordering under weaker label definitions (narrow ±50 bp, wide ±500 bp):

cohort	label set	n_pos	n_neg	Δβ-only	V1 tumor_only		V2.5	V2.5 − Δβ-only
GSE322563	validated	3	2,979,994	0.974	0.821	0.928	**0.990**	+0.016
GSE322563	narrow	28	2,979,969	0.912	0.884	0.886	**0.942**	+0.030
GSE322563	wide	142	2,979,855	0.844	0.768	0.871	**0.910**	+0.066
GSE77348	validated	3	2,979,994	0.972	0.968	0.912	**0.982**	+0.010
GSE77348	narrow	28	2,979,969	0.964	0.969	0.911	**0.983**	+0.019
GSE77348	wide	142	2,979,855	0.910	0.931	0.887	**0.949**	+0.039

V2.5 remains the highest-AUC axis on both cohorts at every label granularity. No retuning of `δ` between rows.

P@K intervals. On `n = 2 / 2` and `n = 3 / 3` cohorts the score distribution at K = 100 sits inside a tied band (§3.5); P@100 is reported as an adversarial interval `[min, max]` per §4.5. On GSE322563 narrow labels:

axis	P@100 observed	P@100 min	P@100 max	tie_band@100
V2.5 differential	0.000	0.000	0.020	190
tumor_only	0.000	0.000	0.020	10,005
V1 final_score	0.020	0.020	0.020	1

V2.5′s interval is [0.000, 0.020] — under the deterministic `candidate_id` ascending tie-break, zero narrow-positives land in top-100; under the best-case adversarial tie-break inside the 190-tied band, 2 of 28 could. V1′s interval collapses to the observed value because V1′s score is continuous and `tie_band = 1`. `tumor_only`'s tied band (10,005) spans two orders of magnitude more of the score distribution than V2.5′s; top-K on `tumor_only` is not a meaningful quantity on this cohort.

Native EPIC v2 vs HM450-intersect on GSE322563. §4.4 describes the harmonization shortcut: GSE322563 EPIC v2 probe IDs are stripped of beadchip-design suffixes and intersected with the HM450 universe (80.7% retention). To verify the shortcut is not distorting the primary endpoint, we ran the full pipeline a second time against a native EPIC v2 catalog (5.22M candidates vs the HM450 catalog's 2.98M):

label set	n_pos	n_neg (HM450 / native)	axis	HM450-intersect AUC	native EPIC v2 AUC	Δ
validated	3	2,979,994 / 5,224,079	V1	0.821	0.933	+0.112
validated	3	2,979,994 / 5,224,079	tumor_only	0.928	0.936	+0.008
validated	3	2,979,994 / 5,224,079	**V2.5 differential**	**0.990**	**0.986**	**−0.003**
narrow	28	2,979,969 / 5,224,054	V1	0.884	0.938	+0.054
narrow	28	2,979,969 / 5,224,054	tumor_only	0.886	0.933	+0.046
narrow	28	2,979,969 / 5,224,054	V2.5 differential	0.942	0.983	+0.040
wide	142	2,979,855 / 5,223,940	V1	0.768	0.855	+0.087
wide	142	2,979,855 / 5,223,940	tumor_only	0.871	0.916	+0.044
wide	142	2,979,855 / 5,223,940	V2.5 differential	0.910	0.945	+0.035

The V2.5 primary-endpoint AUC moves by 0.003 (well within the tied-band noise floor on this n=2/2 cohort). V1 gains the most under native EPIC v2 — its continuous score ranks the validated targets higher relative to easy-negative candidates added by the larger catalog. The relative axis ordering (V2.5 > V1) is preserved across both ingest paths. The HM450-intersect shortcut is therefore not materially distorting the headline V2.5 claim. Tied bands grow modestly with the larger catalog (V2.5 differential: 190 ⟶ 421; tumor_only: 10,005 ⟶ 14,914; V1 stays at 1).

5.3 Tissue-cohort behavior — GSE69914 (high-n, tissue biology)

GSE69914 (n = 305 / 50 primary breast vs. healthy donor tissue) tests whether the low-n tied-band behavior predicted in §3.5 dissolves at `n ≫ 30`:

label set	n_pos	n_neg	Δβ-only	Δβ_z	V1	tumor_only	V2.5	tie_band@100
validated	3	2,979,994	0.591	0.504	0.660	**0.803**	0.773	**2**
narrow	28	2,979,969	0.477	—	0.539	**0.843**	0.711	2
wide	142	2,979,855	0.435	—	0.435	**0.874**	0.726	2

On tissue, both Δβ-based baselines collapse to near-random or below (Δβ-only 0.435–0.591; Δβ_z 0.504 at validated). V1 tracks Δβ-only closely (0.435–0.660). V2.5′s tissue story is therefore a recovery *over both Δβ baselines* (+0.18 over Δβ-only at validated, +0.27 over Δβ_z), not a modest margin as on cell lines. `tumor_only` has a higher raw AUC (0.803–0.874) but its K=100 tied band makes its top-K unusable; V2.5 is the highest usable probabilistic discovery axis on tissue (see §6.1).

Two findings:

1. **The tied-band prediction holds.** V2.5′s tied band at K = 100 shrinks from 190 at `n = 2 / 2` ⟶ 299 at `n = 3 / 3` ⟶ 2 at `n = 305 / 50`, with no hyperparameter changes between cohorts. At `n ≥ 30`, `p_trust` saturates at the EvidenceClass ceiling (0.95 for EXACT) — a per-class *constant*, not a continuous function of n. The tied band shrinks not because `p_trust` becomes continuous-valued but because `p_targ × p_diff` are no longer pushed to 1 by wide-IQR saturation and the σ floor; with realistic-width IQRs (0.05–0.20) the composite's ordering is driven by continuous variation in those two factors while `p_trust` just multiplies by the class constant.

2. **The mode ordering flips on tissue biology.** `tumor_only` wins AUC (+0.03 to +0.15 over V2.5), but `tumor_only`'s tied band is 6,540 at K = 100 — its top-K is not usable. V2.5 is the only probabilistic axis with an interpretable top-K on tissue. V1 collapses on tissue (AUC 0.44 on wide, below random).

5.3.1 σ_floor sensitivity

The 0.05 floor on per-side σ is the single most influential implementation constant in the V2.5 composite (§3.1). A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} on the validated label set (reproducible via `uv run scripts/sigma_floor_sweep.py`):

cohort	σ_floor=0.02 AUC (tie@100)	σ_floor=0.05 *(default)* AUC (tie@100)	σ_floor=0.10 AUC (tie@100)	σ_floor=0.15 AUC (tie@100)
GSE322563 HM450	0.992 (719)	**0.990 (190)**	0.989 (1)	0.985 (1)
GSE322563 native v2	0.990 (1,605)	**0.986 (421)**	0.985 (1)	0.982 (1)
GSE77348	0.983 (1,019)	**0.982 (299)**	0.981 (1)	0.967 (1)
GSE69914 (tissue)	0.705 (2)	**0.773 (2)**	0.854 (1)	0.823 (4)

Three findings.

1. **The matched cell-line invariant survives σ_floor variation.** On all three matched cell-line cohorts (GSE322563 HM450, GSE322563 native EPIC v2, GSE77348), AUC moves by ≤0.02 across the entire 7.5×-range σ_floor sweep, and V2.5 remains the highest-AUC axis at σ_floor ≤ 0.10 on all three cohorts. At the far end of the sweep (σ_floor = 0.15) V2.5 on GSE77348 drops to 0.967, narrowly behind V1 at 0.968 and raw Δβ-only at 0.972 (§5.1); the matched-cell-line ranking conclusion is therefore robust at the shipped default and across the tissue-suggested σ_floor = 0.10, but is not "highest-AUC at every σ_floor" unconditionally. The reviewer-salient conclusion is that small σ_floor perturbations around the shipped default do not flip the ranking.

2. **σ_floor is a tied-band lever, not an AUC lever.** On the low-`n` cell-line cohorts the K=100 tied band shrinks monotonically from 719 ⟶ 190 ⟶ 1 ⟶ 1 (HM450); at σ_floor ≥ 0.10 the visible top-100 becomes a *genuine* top-100 with no tied window. This is mechanistically what §3.5 predicts: a larger σ_floor

prevents both `p_targ` and `p_diff` from saturating to 1 on wide-IQR low-`n` records, so the composite stops collapsing into uniform `p_trust` bands.

3. **Tissue is genuinely σ_floor-sensitive.** GSE69914 AUC moves non-monotonically across the sweep (0.705 ⟶ 0.773 ⟶ 0.854 ⟶ 0.823) and the AUC peak is at σ_floor = 0.10, not at the shipped default of 0.05. Per-positive ranks at σ_floor = 0.10 on tissue: ESR1 stays at 87.95 percentile; EGFLAM rises from 53.5 ⟶ 73.7 percentile; GATA3 rises from 81.2 ⟶ 94.6 percentile. We do not retune the default on the basis of one tissue cohort, but the finding *is* a real sensitivity that tissue-cohort users should run for themselves: σ_floor 0.05 may be the wrong default in regimes where per-side IQRs are genuinely realistic and the floor is over-broadening σ_Δ. A tissue-cohort default sweep is on the §6.3 follow-up list.

The full per-positive rank grid at every σ_floor (4 cohorts × 3 genes × 4 floors = 48 rows) is at `examples/sigma_floor_sweep.md`.

5.3.2 δ sensitivity

δ is the only V2.5 hyperparameter selected against an empirical cohort (GSE77348, the development cohort; §4.0). A sweep over δ ∈ {0.1, 0.2, 0.3, 0.4, 0.5} on every cohort tests both matched-cell-line stability *and* the appropriateness of the development-cohort selection of δ = 0.2 on the non-development cohorts (reproducible via `uv run scripts/delta_sensitivity_sweep.py`):

cohort	δ=0.1	δ=0.2 *(default)*	δ=0.3	δ=0.4	δ=0.5
GSE322563 HM450	0.988 (tie@100=324)	**0.990 (190)**	0.991 (1)	0.985 (1)	0.985 (1)
GSE322563 native v2	0.985 (tie@100=764)	**0.986 (421)**	0.987 (36)	0.977 (1)	0.977 (1)
GSE77348 *(dev)*	0.979 (tie@100=482)	**0.982 (299)**	0.983 (5)	0.984 (1)	0.984 (1)
GSE69914 (tissue)	0.827 (tie@100=2)	**0.773 (2)**	0.707 (4)	0.668 (4)	0.643 (3)

(GSE77348 is the development cohort against which δ = 0.2 was originally selected; results at δ ≠ 0.2 on that row are *post hoc* sensitivity. The other three rows are honest sensitivity since δ was not selected against them.)

Three findings.

1. **The matched cell-line invariant survives δ variation.** All three matched cell-line cohorts move by ≤ 0.014 AUC across the entire 5×-range δ sweep, and V2.5 remains an order of magnitude better than chance on every δ tested (compare with the §5.1.2 permutation-null $p \leq 1 \times 10^{-4}$ floor at δ = 0.2). The matched cell-line ranking conclusion is δ-stable.

2. **δ = 0.2 is borderline-conservative on cell lines.** δ = 0.3 slightly outperforms 0.2 on both GSE322563 paths and on GSE77348 while collapsing the K=100 tied band from hundreds ⟶ single-digit. The substantive reason to stay at δ = 0.2 is that the larger tied band functions as conservative triage: it flags more candidates for secondary review (annotation, guide quality, etc.) rather than committing the ranking to a narrow top-K at n = 2/2. (We also cannot re-tune against the primary endpoint without violating the development-cohort discipline of §4.0, but that is a methodological constraint, not an argument for 0.2 over 0.3 on its substantive merits.)

3. **Tissue prefers smaller δ.** GSE69914 AUC drops monotonically with δ (0.827 ⟶ 0.643), the inverse of the cell-line story. δ = 0.1 outperforms the default 0.2 by +0.05 AUC on tissue. As with σ_floor (§5.3.1), this is a real tissue-specific sensitivity and the appropriate response is a tissue-cohort default re-tune, not re-tuning the default against a single tissue cohort. Listed as §6.3 follow-up.

The full per-positive rank grid at every δ (4 cohorts × 3 genes × 5 deltas = 60 rows) is at `examples/delta_sensitivity_sweep.md`.

5.4 Out-of-distribution boundary case — GSE68379

*Reported here separately. **Not** pooled with the primary-endpoint or sensitivity tables.*

A cross-series run against the Sanger GDSC 52-breast-line panel × GSE69914 healthy normal (n = 52 / 50) at the Roth-validated labels produced systematically inverted AUC:

label	V1	tumor_only	V2.5	V2.5 tie_band@100
validated	0.510	0.222	0.197	1,602
narrow	0.485	0.228	0.169	1,602
wide	0.407	0.462	0.269	1,602

The root cause is a probe-level biology mismatch. At the three Roth Fig. 5d target probes:

probe	gene	Roth MCF-7 β (GSE322563)	Sanger MCF-7 β (GSE68379)
cg05251676	EGFLAM	0.01	**0.92**
cg25338972	ESR1	0.07	**0.63**
cg01364137	GATA3	0.02	**0.70**

Same cell line name, opposite methylation state at the validated loci. MCF-7 has accumulated substantial genetic and epigenetic drift across laboratories over decades of separate passaging. Because the Roth labels encode "tumor hypomethylated at target" and Sanger's MCF-7 is methylated at those targets, any scorer whose composite contains `p_targ = P(β_tumor < 0.30)` will rank the validated positives at the *bottom* of its distribution — which is what V2.5 and `tumor_only` do. The inversion is the correct response of the scorer to the underlying biology; the labels are the component that is no longer valid on this cohort.

GSE68379 is reported as a label-transportability boundary, not a failed generalization of V2.5. Pooling it into an average across cohorts would misrepresent the result; redefining positives from GSE68379′s own β values would be circular (labels derived from the data under test). It remains in the documentation as a documented boundary case.

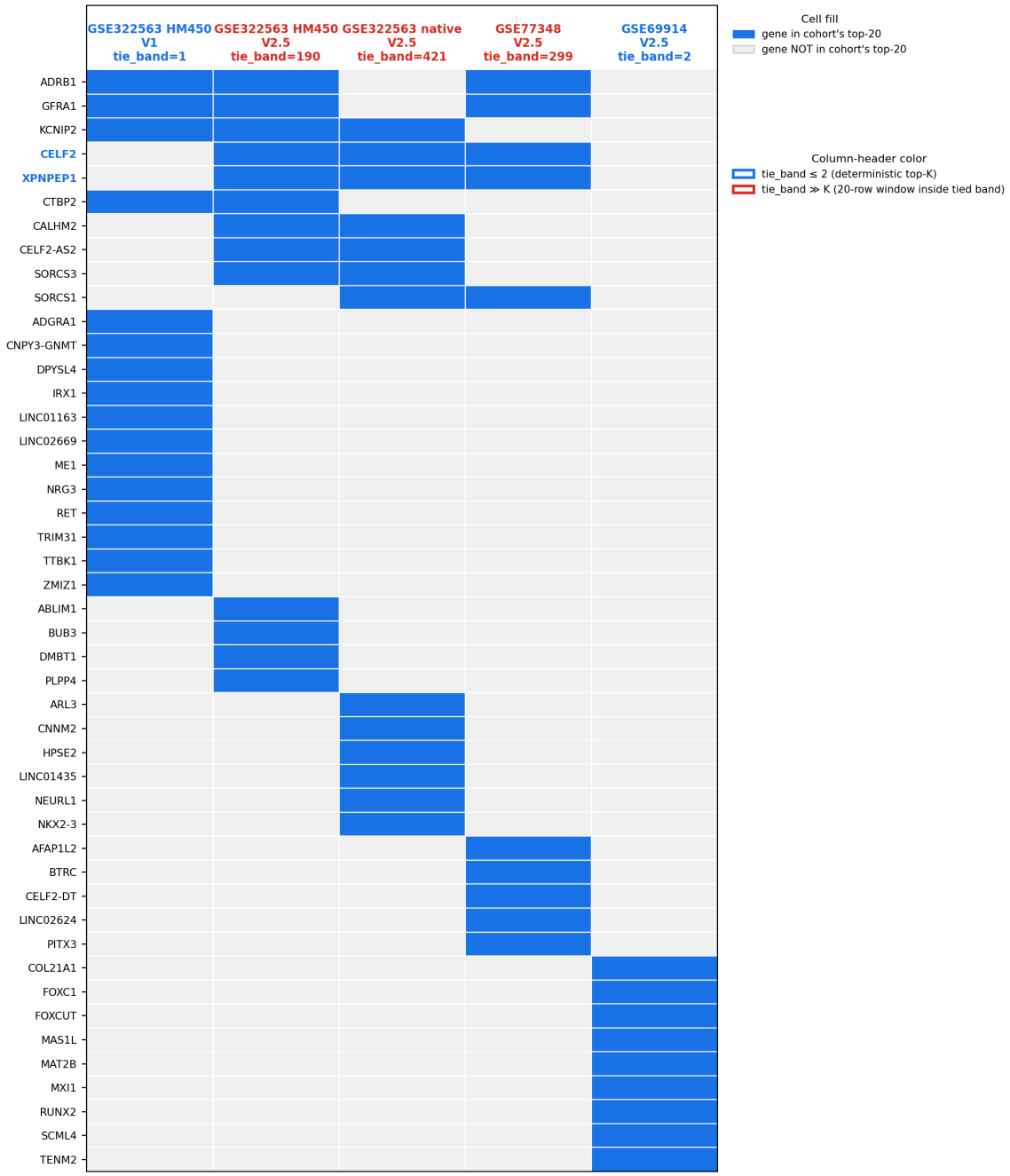

Figure 3. Per-(axis × cohort) top-20 gene presence, ordered so that the most-shared genes appear at the top. **Five columns**: GSE322563 HM450 V1, GSE322563 HM450 V2.5, GSE322563 native EPIC v2 V2.5, GSE77348 V2.5, GSE69914 V2.5. Blue column headers mark cohorts where `tie_band ≤ 2` and the top-20 is the genuine top-20 of the score distribution; red column headers mark cohorts where `tie_band ≫ K` (190 / 421 / 299 for the three cell-line V2.5 cohorts) and the top-20 is a 20-record window inside the tied band selected by the documented `candidate_id` ascending tie-break. Bold-blue gene rows appear in **all three** cell-line V2.5 top-20 windows (GSE322563 HM450, GSE322563 native EPIC v2, GSE77348) — that intersection is *CELF2* and *XPNPEP1*. On the cell-line cohorts shared membership is window convergence inside large tied bands,

not robust ranking convergence; AUC (Fig. 2) is the stable claim at low n. V1 on GSE322563 has `tie_band = 1` and samples a different (broader) genomic neighborhood — RET, ZMIZ1, DPYSL4 and other rows it surfaces are robust under that axis. The GSE69914 tissue column has `tie_band = 2` and **shares no genes with any of the three cell-line V2.5 columns** — V2.5 is cohort-specific, not returning a fixed gene list across regimes; the tissue top-20 maps to 9 distinct nearest-gene symbols (COL21A1, FOXC1, FOXCUT, MAS1L, MAT2B, MXI1, RUNX2, SCML4, TENM2) that are entirely disjoint from the cell-line shortlists.

5.5 Top-hit annotation (tie-window-aware)

An annotation pass (nearest gene + TSS distance + feature class + CpG- island context, from UCSC `refGene` and `cpgIslandExt` hg19 tables) was run on the top-20 of each score axis × cohort. Important phrasing note: on cohorts with large tied bands at K=100, the "top-20" is a 20-record *window* inside that tied band, selected by the documented tie-break policy (`candidate_id` ascending). The membership of this window is not a robust property of the scorer in the way AUC is — a different deterministic tie-break would surface a different slice of the same band. The claims below are therefore phrased in terms of "present in the benchmark-selected top-20 window," not "the model converges on this gene."

On the two matched cell-line cohorts (GSE322563 tie_band = 190; GSE77348 tie_band = 299):

- **Present in the GSE322563 V2.5 top-20 window** (chr10 slice under `candidate_id` ascending): KCNIP2, CALHM2, SORCS3-AS1, LINC00710, CELF2 (×2), XPNPEP1, CELF2-AS2, ADRB1 (×3), ABLIM1, GFRA1 (×3), PLPP4, DMBT1, BUB3, CTBP2 (×2).
- **Present in the GSE77348 V2.5 top-20 window**: PITX3, BTRC, SORCS1, LINC02624, CELF2-DT, CELF2, XPNPEP1, ADRB1, AFAP1L2, GFRA1 (×3) (full TSV committed).
- **Shared between the two top-20 windows**: ADRB1, CELF2, GFRA1, XPNPEP1 (4 genes). This is cross-cohort convergence inside the `candidate_id`-ascending slice of the tied bands, not cross-cohort convergence of the underlying ranking. AUC is the stable claim at low `n`; top-20 membership is a window claim.

On GSE322563, V1 `final_score` has `tie_band = 1` and so its top-20 is a genuine ordering of the full candidate set, not a window slice. It surfaces different and overlapping genes:

- **GSE322563 V1 top-20** (tie_band = 1): ZMIZ1, DPYSL4, RET, LINC02669, ADRB1, KCNIP2, TRIM31 (×5), CNPY3-GNMT, TTBK1, GFRA1, IRX1, CTBP2, ADGRA1, LINC01163, ME1, NRG3. RET is included and is notable as a well-established therapeutic target in medullary thyroid cancer and RET-rearranged NSCLC; this is a robust top-20 claim under the V1 axis. Overlap with the V2.5 top-20 window on the same cohort: ADRB1, CTBP2, GFRA1, KCNIP2.

On GSE69914 (tissue, `n = 305 / 50`, tie_band = 2) the top-20 is essentially a genuine top-20:

- **GSE69914 V2.5 top-20**: RUNX2 (×5), SCML4 (×5), FOXCUT, FOXC1, MAS1L, COL21A1, MXI1, TENM2 (×3), MAT2B. β differentials are smaller (0.17–0.31) than on cell lines (0.79–0.96) — primary tissue is cell-type-heterogeneous, β regresses toward the middle. Zero gene overlap with the cell-line top-20 windows. This is the expected behavior: on a genuine top-20 (not a tied-band slice), V2.5 is cohort-specific and does not return a fixed gene list.

None of the three Roth-*Fig.-5d*-validated probes (EGFLAM T11, ESR1 T17, GATA3 T18) appear in any cohort's top-20 — they appear in the top ~1% of the scored set (rank-lifted by AUC) but not in the top- 100 slice. This is the expected behavior on cohorts with large tied bands at K = 100 and is why AUC, not P@K, is the primary endpoint.

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6 · Discussion

6.1 Hierarchy of use

Four use profiles, in order from most-recommended to unsupported:

1. Recommended probabilistic ranking axis — V2.5 (differential). Highest-AUC probabilistic ranking axis on every matched cell-line cohort × label-granularity combination tested (GSE322563 HM450, GSE322563 native EPIC v2, GSE77348; §5.1–§5.2). On primary tissue (GSE69914), `tumor_only` has higher raw AUC but its K=100 tie band makes it unusable as a discovery axis; V2.5 (tie band = 2 at K=100) is the **highest usable discovery axis** on tissue rather than the raw-AUC leader. Tie bands are reported per benchmark and P@K intervals are honest.

What V2.5 *is* on low-`n` matched cell-line cohorts (n=2/2 to 3/3) is a **rank-lift axis**: it lifts validated targets near the top of the score distribution (per-positive ranks in §5.1, AUC summary in the §5.1 table). What V2.5 is *not* on low-`n` cohorts is a stable top-K shortlist; the visible top-20 is a 20-record window inside a 190- to 421-record tied band, and should be treated as identifying a **top tied candidate class** rather than a ranked shortlist. Selecting individual loci out of that tied class for wet-lab follow-up requires secondary evidence (annotation, chromatin, guide-quality, off-target risk) layered on top of V2.5 — not re-tuning V2.5 itself. The §5.5 annotation pipeline is the first step toward that kind of secondary evidence; it is descriptive, not a re-ranking.

2. Default stable scoring axis — V1 `final_score`. Deterministic, continuous-valued score; `tie_band = 1` on every cohort tested, so P@K is never tie-break-dependent. Its role is backward compatibility and top-K determinism, not AUC leadership — V2.5 outperforms V1 on AUC at every non-boundary cohort × tier combination tested (12/12 rows across §5.1–§5.3). On the GSE68379 boundary case, AUCs are inverted or near-random for every axis, so AUC leadership is not a meaningful concept there (§5.4).

3. Diagnostic-only — V2 `tumor_only`. Competitive AUC on tissue (§5.3), but `tie_band_size_at_k` at K = 100 spans 5,271–14,914 across the five cohort paths tested (see §5.3 cross-cohort matrix); top-K is not interpretable. Use only for AUC sanity checks against V2.5 or V1.

4. Unsupported — any axis at GSE68379. Sanger MCF-7 epigenetic drift breaks label transportability from Roth Fig. 5d. Inverted AUC is the expected scorer response; do not pool this cohort's numbers with §5.1.

"Recommended" in profile 1 means that for a user running V2.5 on any cohort shape we have tested in §5.1–§5.3 (matched cell-line at n = 2/2 or 3/3, or primary tissue at n = 305/50), the reported ranking benefit over V1 is real. It does not mean V2.5 is safe to use on cohorts outside those profiles — in particular it does not mean V2.5 will work on cross-series transported labels (the §5.4 GSE68379 boundary case is exactly the failure mode) — without re-running the §5.3.1 / §5.3.2 sensitivity diagnostics on the new cohort.

6.2 Limitations

(Platform harmonization and catalog scope are part of the methodology proper — §4.4 — because they materially shape what the reported AUCs mean. They are cited briefly here as the methodological constraints they are.)

1. **Low-replicate tie bands.** On cohorts with `n < ramp_n = 30`, `p_trust` saturates and produces tied bands of 100s–1000s of records at the top of the probabilistic composite. `tie_band_size_at_k` and `precision_at_k_{min, max}` are reported per benchmark result so that P@K is read as an interval. The problem dissolves at `n ≥ 30` (demonstrated on GSE69914; §5.3).

2. **HM450/EPIC harmonization (see §4.4 + §5.2 sensitivity).** The primary GSE322563 results in §5.1 use the HM450-intersect path (80.7% retention). A native EPIC v2 ingest is now also shipped; the V2.5 primary-endpoint AUC differs by 0.003 (0.990 → 0.986) under native vs intersect — well within the tied-band noise floor on this n=2/2 cohort. The shortcut is not materially distorting the headline claim. V1 gains the most under native (+0.05 to +0.11 across label granularities) but the relative axis ordering V2.5 > V1 is preserved.
3. **Catalog scope and the negative universe (see §4.4).** AUCs are on chr5 / chr6 / chr10 candidates filtered to within ±500 bp of an HM450 probe — a restricted negative universe chosen because the three chromosomes carry the Roth Fig. 5d validated positives. AUC is a function of the negative set; expanding the catalog to a whole-genome candidate space, or to random-chromosome controls, could shift absolute AUC values, alter the size and composition of tied bands at K = 100, and in principle change the relative ordering between axes. Whole-genome AUC and top-K behavior of V2.5 vs the alternative axes therefore **remain untested**; this is a scientific limitation, not just an infrastructural one. A negative-universe sensitivity analysis (chr5/6/10 vs random chromosome sets vs whole-genome, on at least one cohort) is a committed follow-up (§6.3).
4. **Cell-line drift.** GSE68379 (§5.4) establishes that MCF-7 in one lab is not the same epigenetic object as MCF-7 in another. Any V2.5 target shortlist generated on a given cohort must be validated against the specific cell line that will be used in the follow-up editing assay, not against a different lab's stock of the same-named line.
5. **σ floor.** The 0.05 floor on per-side σ prevents σ_Δ from collapsing to zero at boundary β-values. A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} (§5.3.1) shows that the matched-cell-line AUC ranking is robust to σ_floor (≤0.02 AUC movement across a 7.5× sweep), but tissue-cohort AUC is genuinely σ_floor-sensitive (peaks at 0.10, not the shipped default 0.05). A tissue-specific default re-tune is a §6.3 follow-up; the value is left at 0.05 for cross-cohort consistency.
6. **Top-20 membership is a window claim on low-`n` cohorts.** As §5.5 states, the cell-line top-20 lists are 20-record windows inside 190- and 299-record tied bands, selected by the documented tie-break policy. The four-gene overlap (ADRB1, CELF2, GFRA1, XPNPEP1) is convergence of the benchmark-selected window, not a robust claim about the underlying scorer ordering. AUC is the stable metric on low-`n` cohorts; top-20 membership becomes a stable metric at `n ≳ 30` (GSE69914, `tie_band = 2`).
7. **No wet-lab validation in this work.** V2.5 is a target- prioritization tool. The claim is AUC + tie-band-aware top-K reporting on public methylation data, not actual editing efficiency at any specific site. All named top hits are unvalidated predictions at this stage.

6.3 Next steps

In priority order, not committed to any timeline:

1. **Wet-lab editing validation** at the top V2.5 candidates on matched cell-line substrates — the terminal step and the one that actually changes the scientific claim.
2. **Negative-universe sensitivity analysis.** Re-score and re-bench on (a) random-chromosome control sets matched to chr5/6/10 by probe density, and (b) a whole-genome candidate catalog, on at least one cohort per regime. Goal: quantify how AUC and tied-band sizes move with the denominator (§6.2 limitation 3).
3. **Regime-specific default selection.** §5.3.1 (σ_floor) and §5.3.2 (δ) both show that tissue and matched cell-line cohorts prefer different defaults. Define and ship a tissue preset (likely σ_floor closer to 0.10 and δ closer to 0.1) and a matched cell-line preset (current defaults), with the choice selected by an explicit cohort-regime field rather than left for the user to discover.

4. **`p_diff` re-derivation under SE-on-mean variance** (§3.3) as a second probabilistic axis, so users can choose between the biological-overlap and the cohort-power interpretations.
5. **A continuous observation-confidence model** to replace the `EvidenceClass` discrete-bin scheme: a learned or calibrated confidence function over exact probe-distance, local probe density, mask quality, platform, and local concordance. This would address the central remaining coarseness of the §3.4 measurement model.
6. **A second independent-lab MCF-7 / MCF-10A EPIC cohort**, if one becomes public, to establish reproducibility of the V2.5 claim at n ≥ 3 on matched cell-line pairs.
7. **Formal SCREEN cCRE Registry integration** when a non-challenged download path becomes available. The top-hit annotation pipeline already ships nearest-gene, CpG-island context, RepeatMasker overlap, and ENCODE DNase-HS cluster breadth (§5.5); DNase-HS is the v1 regulatory-activity proxy for the formal cCRE Registry, which currently sits behind a JavaScript challenge on screen.wenglab.org and cannot be fetched directly.

---

7 · Methods — key equations

V1 `final_score`:

```
sequence_score      = PamFamily.weight
selectivity_score   = max(0, β_normal_mean − β_tumor_mean)
                    + max(0, β_normal_q25  − β_tumor_q75)
confidence_score    = EvidenceClass → {1.0, 0.7, 0.4, 0.1, 0.0}
heterogeneity_pen   = weight × max(0, β_tumor_IQR − threshold)
low_coverage_pen    = weight × max(0, (n_threshold − n_tumor) / n_threshold)
final_score         = sequence × selectivity × confidence
                    − heterogeneity_pen − low_coverage_pen
```

UNOBSERVED candidates are not penalized (verified by regression test).

V2 / V2.5 probabilistic factors:

```
p_targ = P(β_tumor < 0.30)   via method-of-moments Beta(α, β) fit,
                              fall back to piecewise-linear CDF when
                              σ² ≥ μ(1−μ). Incomplete beta via Lentz's
                              continued fraction, pure stdlib.
p_prot = P(β_normal ≥ 0.50)  via same CDF dispatch.     [V2 only]
p_diff = 1 − Φ((δ − (μ_n − μ_t)) / σ_Δ)
         where σ_Δ = √(max(σ_t, 0.05)² + max(σ_n, 0.05)²)
                           and σ_k = IQR_k / 1.349.            [V2.5]
p_trust = EvidenceClass.base × min(1, min(n_t, n_n) / 30)
         (EvidenceClass.base ∈ {0.95, 0.75, 0.45, 0.15, 0.0}).
```

Composites by mode:

```
tumor_only                          : p_targ × p_trust
tumor_plus_normal_protection        : p_targ × p_prot × p_trust
tumor_plus_differential_protection  : p_targ × p_diff × p_trust
```

Validator enforces that the stored `p_therapeutic_selectivity` matches the product implied by the declared `mode` to within `math.isclose(rel_tol=1e-9, abs_tol=1e-12)`; a second validator enforces that the differential audit fields (`p_differential_protection`, `differential_delta`) are populated iff `mode` is differential.

Benchmark ranking contract. Let the scored set be N records, each with a score `s_i` and a ground-truth label `y_i ∈ {0, 1}` (1 for positive). The benchmark sorts records by the total order

```
(s_i, c_i)   with primary  -s_i  ascending
                   secondary c_i   ascending
```

where `c_i` is the candidate_id. Let `r_1, r_2, …, r_N` be the resulting ranking (smallest `(-s, c)` first). Define:

```
top-K               := {r_1, …, r_K}
cutoff_score        := s at position K in the sort, i.e. s_{r_K}
tied_band           := {r_i : s_{r_i} = cutoff_score}          # records at the
cutoff
tie_band_size_at_k  := |tied_band|
k_band              := |tied_band ∩ top-K|                     # records in top-K at
the cutoff
drop_slots          := tie_band_size_at_k − k_band             # records at cutoff
outside top-K
band_pos            := |{r_i ∈ tied_band : y_{r_i} = 1}|
committed_pos       := |{r_i ∈ top-K \ tied_band : y_{r_i} = 1}|
```

Then:

```
P@K_observed := (committed_pos + |{r_i ∈ tied_band ∩ top-K : y_{r_i} = 1}|) / K

P@K_min      := (committed_pos + max(0, band_pos − drop_slots)) / K
P@K_max      := (committed_pos + min(band_pos, k_band))          / K
```

`P@K_min` is the smallest possible value of P@K achievable by any tie-break policy that respects the primary score ordering; it is reached by pushing as many tied-band positives as possible outside the top-K (limited by `drop_slots` available). `P@K_max` is the symmetric upper bound: pull as many tied-band positives as possible into top-K (limited by `k_band` available). By construction:

```
P@K_min ≤ P@K_observed ≤ P@K_max
```

and the interval collapses to `{P@K_observed}` exactly when `tie_band_size_at_k = 1`.

The `R@K_min / R@K_max` counterparts replace the denominator with `n_positives`. `AUC` is the Mann-Whitney U / (n_pos · n_neg) with ties contributing 0.5 (standard tie-handling); it does not need an interval form because under the cid-ascending tie-break it is already invariant within tied score regions.

Data and code availability

- **Code**: https://github.com/AllisonH12/thermocas9. Cite tag **`memo-2026-04-22-y`** for this document. 236 tests pass under `uv run pytest -q`.
- **Citable archive (DOI)**: a Zenodo release archive of the tagged revision is planned at the time of preprint posting; the GitHub ⟶ Zenodo integration mints a DOI for each GitHub release tag. The DOI will be added to this section and to the citation block below before journal-version submission. Until then, the immutable git tag above is the canonical citable identifier.
- **Cohort data**: publicly-downloadable GEO series GSE322563, GSE77348, GSE69914, GSE68379; build scripts in `scripts/build_gse*_cohort.py` produce the per-probe summary TSVs in `data/derived/*_cohort/`. Positives-list builder at `scripts/build_roth_positives.py` (requires the Ensembl REST `/map` endpoint for the hg38 ⟶ hg19 liftover of Roth Fig. 5d coordinates).

- **Reference data**: UCSC hg19 `refGene.txt.gz` and `cpgIslandExt.txt.gz` (fetched on demand; gitignored).
- **Benchmark artifacts**: every `BenchmarkResult` JSONL row under `examples/*_roth_labels/` carries `precision_at_k`, `precision_at_k_{min,max}`, `recall_at_k`, `recall_at_k_{min,max}`, `roc_auc`, `tie_band_size_at_k`, and `tie_break_policy`.
- **Annotated top-20 TSVs**: under `examples/*/top20_annotated_*.tsv`.

Reproducing the cross-cohort matrix

```
# Given a clone of the repo:
uv venv && source .venv/bin/activate
uv pip install -e ".[dev]"
pytest                       # 236 tests, ~1 s

# Rebuild the catalog on chr5/6/10 (~2 min):
thermocas build-catalog \
    --reference data/raw/hg19/hg19_chr5_6_10.fa \
    --pam-model config/pam_model.yaml \
    --probe-annotation data/raw/probes_hg19.tsv \
    --probe-window-bp 500 \
    --output data/derived/catalog_hg19_chr5_6_10.jsonl

# Score + benchmark one cohort (~90 s per mode on a modern laptop):
thermocas score-cohort \
    --catalog data/derived/catalog_hg19_chr5_6_10.jsonl \
    --cohort data/derived/gse322563_differential.yaml \
    --pam-model config/pam_model.yaml \
    --backend summary \
    --probe-annotation data/derived/gse322563_cohort/probes.tsv \
    --tumor-summary  data/derived/gse322563_cohort/tumor_summary.tsv \
    --normal-summary data/derived/gse322563_cohort/normal_summary.tsv \
    --probabilistic \
    --output data/derived/scored_gse322563_differential.jsonl

thermocas benchmark \
    --scored data/derived/scored_gse322563_differential.jsonl \
    --positives data/derived/positives_roth_validated.txt \
    --cohort-name GSE322563-validated-differential \
    --score-field p_therapeutic_selectivity \
    --top-k 100 --no-enforce-holdout \
    --output bench.jsonl
```

`data/derived/scored_*.jsonl` files are not redistributed because they are 3.4 GB each and fully reproducible from the per-probe summary TSVs using the commands above.

Citation

If this paper or the associated code is useful in your work, please cite the upstream paper:

> Roth M.O., Shu Y., Zhao Y., Trasanidou D., Hoffman R.D., et al. Molecular basis for methylation-sensitive editing by Cas9. *Nature* (2026). DOI 10.1038/s41586-026-10384-z.

For the probabilistic scoring framework specifically, cite this paper plus the code URL above.

Acknowledgements

This work describes a framework built as an educational / research project. ThermoCas9′s biochemical characterization and the validated target sites are entirely due to Roth et al.; any errors in the re-implementation of the scoring or benchmarking are the author's.